



- (b) The table shows part of the instruction set for a processor. The processor has one register: the Accumulator (ACC).

| Instruction                                                                                                                                                                         |           | Explanation                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------|
| Opcode                                                                                                                                                                              | Operand   |                                                                             |
| AND                                                                                                                                                                                 | #n/Bn/&n  | Bitwise AND operation of the contents of ACC with the operand               |
| AND                                                                                                                                                                                 | <address> | Bitwise AND operation of the contents of ACC with the contents of <address> |
| XOR                                                                                                                                                                                 | #n/Bn/&n  | Bitwise XOR operation of the contents of ACC with the operand               |
| XOR                                                                                                                                                                                 | <address> | Bitwise XOR operation of the contents of ACC with the contents of <address> |
| OR                                                                                                                                                                                  | #n/Bn/&n  | Bitwise OR operation of the contents of ACC with the operand                |
| OR                                                                                                                                                                                  | <address> | Bitwise OR operation of the contents of ACC with the contents of <address>  |
| <address> can be an absolute or a symbolic address<br># denotes a denary number, e.g. #123<br>B denotes a binary number, e.g. B01001010<br>& denotes a hexadecimal number, e.g. &4A |           |                                                                             |

Write the bit manipulation instructions that can be used to set only the most significant bit to 1 in an 8-bit register.

The instructions need to work on a register that contains any 8-bit binary number.

.....  
 ..... [2]

